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Executive Summary

Silver is produced in modest volumes (~26–28 kt/year) but has large global reserves (≈0.6 million t). Current mine output (~27 kt in 2024) is led by Mexico (~8 kt), Peru (~3.5 kt), China (~3.4 kt), the United States (~1.05 kt), Bolivia (~1.5 kt) and Australia (~1.0 kt)【19†L3991-L3994】【19†L4001-L4002】. Silver reserves are heavily concentrated: USGS (2025) estimates Peru (110 kt), Russia (92 kt), Australia (91 kt) and Poland (59 kt) hold over half the world's silver【76†L52-L60】【48†L52-L60】. Most silver today is obtained as a by-product of base-metal and gold mining – especially from lead–zinc and copper ore bodies – with only ~25–30% from primary silver deposits【51†L169-L174】. Major deposit styles include epithermal veins (e.g. Mexico, Peru), volcanogenic massive sulphides and carbonate-replacement ("CRD") deposits, and porphyry-related systems (often copper or gold mines with Ag credits). High-grade silver (>200–500 g/t) typically occurs in epithermal vein-rich districts (e.g. Fresnillo, Mexico) and CRD–VMS areas (e.g. Keno Hill, Yukon).

In the UK, silver occurs chiefly as low-grade silver in lead–zinc veins. Historic production (e.g. Combe Martin, Devon and lead mines of Derbyshire/North Pennines) supplied silver, but today the UK has no operating silver mines and very minor output (~0.63 t in 2023【19†L3967-L3969】). Exploration is limited to small VMS/vein targets (e.g. Cononish gold–silver veins) and the remoteness or low grades of targets make projects uneconomic under current prices and regulatory regimes. UK mining faces stringent environmental/permitting requirements, high costs and competing land uses, meaning large-scale silver development is unlikely without major price or policy changes.

Below we detail global production/reserves trends, dominant deposit types, and prospective regions for high-grade silver, followed by a focused UK section (geology, history, permitting, economics). Key uncertainties include data gaps on new exploration discoveries and evolving market/regulatory factors (e.g. recycling, green demand).

Global Production and Reserves

Worldwide silver mine production has been stable at ~26–28 million kg annually (26–28 kt) over the past decade【19†L4001-L4005】. In 2024 output was ≈27.8 kt, slightly up from ~26.7 kt in 2023【19†L4001-L4005】. Production by country is dominated by Mexico (~8,000 t in 2024) and trailing leaders Peru (3,510 t), China (3,424 t), the United States (1,050 t) and Bolivia (1,486 t)【19†L3991-L3994】【19†L4001-L4002】 (Figure). Other notable producers include Australia (~1,045 t), Chile (1,354 t), Poland (1,531 t) and Russia (1,275 t)【19†L3991-L3994】【19†L4001-L4002】. Global production reflects large-scale base-metal mining – e.g. Peru's output is driven by copper/zinc mines with silver credits. Indeed, only ~25–30% of silver comes from primary Ag–Au mines【51†L169-L174】; the rest is recovered from Cu, Pb, Zn and Au mines.

【72†embed_image】 *Figure: Top silver-producing countries (2020–24). Mexico, Peru and China far outpace other nations (data: BGS World Mineral Production).*

Global silver reserves are estimated at ~0.6–0.7 Mt (600–700 kt) (USGS 2026). The largest known reserves are in Peru (~110 kt Ag), Russia (92 kt) and Australia (91 kt)【76†L52-L60】【48†L52-L60】. Poland (59 kt) and China (~67 kt) also hold major reserves (USGS 2026), while Mexico (≈37 kt, reflecting centuries of mining) and the USA (~23 kt) are smaller on the list【48†L52-L60】【82†L52-L60】. These top five

countries account for roughly half of global reserves. Note that reported “reserves” are economic at current costs; total resource figures are larger but less certain. (For reference, 1 kt = 1,000 t; 1 troy ounce≈0.0311 kg.)

Silver Deposit Types and Grades

Silver is found in a range of hydrothermal deposits. **Epithermal veins** (both low- and high-sulphidation) host many high-grade silver deposits. For example, Mexico’s Fresnillo district (low-sulphidation epithermal) has historically produced silver at hundreds of g/t grades【85†L345-L354】. The Juanicipio deposit (Zacatecas, Mexico) has combined indicated resources of ~300 Mt at ~511 g/t Ag【38†L21-L24】. Similarly, Peru’s Santa Rosa epithermal veins yield high-grade Ag–Au. **Volcanogenic Massive Sulphide (VMS)** and **Carbonate Replacement (CRD)** deposits often carry significant Ag with Pb–Zn. Classic examples include Keno Hill (Yukon) and Idaho Coeur d’Alene/Bayhorse (USA), where silver grades often range tens to hundreds of g/t. **Porphyry** and **Cu-Au mines** also contribute to Ag output: many copper/gold porphyries (e.g. Antamina in Peru【85†L352-L358】, Grasberg in Indonesia) yield tens of millions of ounces of silver as a by-product. Indeed, USGS notes polymetallic deposits (e.g. KGHM’s Polish copper belt) contain over two-thirds of world Ag resources【51†L169-L174】.

Typical grades vary by deposit: world-average Ag grades are often <100 g/t. High-grade chunks (“bonanza veins”) can exceed 1,000 g/t locally. For example, narrow silver-bearing veins in Mexico and the American West commonly assay several hundred g/t silver or more. By contrast, bulk-tonnage disseminated ores (e.g. porphyries, zinc mines) may carry 50–200 g/t Ag. Ore bodies can range from thousands of tons (small vein shoots) to hundreds of millions of tons (porphyry deposits like Penasquito, Mexico with ~300 Mt containing 6.6 Moz Ag【85†L293-L300】).

Exploration Indicators: High silver systems often associate with Sb, Hg or Tl anomalies (pathfinder elements in epithermal systems). Alteration minerals like silica, clays and adularia often mark epithermal zones. Lead–zinc sulphides (galena, sphalerite) are cues in CRD/VMS. Geophysical surveys can highlight massive sulphide bodies or buried intrusions. Many silver districts are “hidden” under sediment cover, requiring drilling guided by geochemistry and remote sensing.

Key Regions and Districts

Silver’s richest districts cluster in established mining countries. **Mexico** (“the world’s silver heartland”) hosts multiple epithermal provinces (Zacatecas, Durango, Sonora) with dozens of mines. Key operations include Fresnillo (Fresnillo plc), Peñasquito (Newmont) and Juanicipio (MAG Silver/Fresnillo)【51†L221-L229】. **Peru** and **China** also rank high, but much Ag in Peru comes from Cu–Zn–Ag polymetallic mines (e.g. Antamina, Cerro de Pasco)【51†L185-L194】【85†L352-L358】. **Bolivia** (San Cristóbal mine, 17.6 Moz Ag in 2025【85†L307-L314】) and **Chile** (KGHM’s operations in Poland vs Chile’s older Cerro Moro or Spence Cu) contribute significant silver. **Australia**’s Cannington (BHP, now closed) was a giant Ag–Pb mine【85†L358-L366】, and smaller producers (Perseverance, Mt. Lyell) remain. **United States** has many silver districts: historic Comstock Lode (Nevada), Coeur d’Alene (Idaho), and current producers like Greens Creek (Alaska) and Rochester (Nevada).

Other promising regions include **Argentina** (high-grade epithermal silver–gold like Navidad, Cerro Moro), **Canada** (Keno Hill, Newfoundland VMS), **Russia** (Dukat/Polymetal’s deposits【51†L209-L218】), **Turkey**, and **Central Asia** (Kyrgyzstan’s Kumtor Pb–Zn). Northern and Arctic regions (e.g. Yukon’s Carlin-type Au-Ag) are under exploration. In general, countries with prolific base-metal and gold belts (e.g. Poland, Sweden, Romania, Ireland) often yield silver as a byproduct, even if small-scale.

High-Grade Potential: Regions with known bonanza veins are top targets. Mexico (especially Sierras Madre Occidental and Oriental) consistently yields multi-hundred g/t Ag. The Peruvian Andes and Southern Peru (Puno, Huancavelica) host silver epithermals. Argentina's Santa Cruz and Jujuy provinces, and Bolivia's Potosí, hold large, high-grade veins. In the US, Nevada's Cortez or Golden Triangle (Montana) districts see renewed silver-focused exploration. Emerging targets include underexplored belts in Africa (e.g. foreign-led projects in DRC, Burkina Faso, Morocco with potential Ag credits) and the Himalayan belt (Pakistan, India). **West Africa's** base-metal districts (e.g. Burkina, Cote d'Ivoire) also produce byproduct silver from zinc/lead mines. Overall, highest grades remain in known historic districts; new discoveries may come from concealed extensions of those or analogous geologic settings.

UK Silver Mining: Geology and Projects

The UK has never been a major silver source. Silver occurs in association with lead–zinc vein systems, especially in Carboniferous "base-metal" belts of England and Wales. In England's Mendips and North Pennines, Galena (lead sulphide) carried silver (historically ~150 g/t)【19†L3967-L3969】. The most famous English silver town was Combe Martin (Devon) in the Middle Ages; it yielded thousands of tonnes of ore at ~50–100 g/t Ag for coins and the Crown【57†L5-L9】. Derbyshire's Blue John mines (predominantly fluorite now) once produced handfulls of silver, and Wales (Dolaucothi gold mine) yielded minor Ag. In Northern Ireland, the Cavanacaw and Whitesands lead–silver mines produced ore with ~70–100 g/t Ag. None of these were large by modern standards – typical UK vein widths <1 m, and tonnages often <1 Mt.

Today, **UK silver output is negligible**. Official data report ~0.63 t in 2023 (from minor byproduct recovery)【19†L3967-L3969】, basically de minimis. There are no active primary silver or lead–zinc mines. The only significant precious-metal mine is Cononish (Scotland), focused on gold with very low Ag byproduct (<1 g/t). UK reserves are not reported by USGS (essentially zero economic reserve).

Exploration: A few junior companies have explored UK base-metal prospects. For example, Frongoch area in Wales (the North Pennine Orefield) has prospective veins, and Bolivar Minerals briefly drilled the legacy lead mines. In Scotland, Cononish's veins extend beyond current mine. However, most known targets are small and deeply weathered. No major new discoveries have been announced.

Geology: UK's hydrothermal veins were formed mainly in the Carboniferous (Basins & Blocks of northern England/Wales) and Devonian (Mendips). They typically host Pb–Zn–Ag–Ba–F (e.g. galena+sphalerite+fluorite), rather than massive auriferous or polymetallic bodies. As such, grades are modest (hundreds ppm Ag). The Permian of Cornwall had Sn–Cu veins (some Ag and Au) but nothing large-scale.

Permitting & Economics: The UK has a mature and stringent mining/ environmental regime. Any new mine must navigate land rights (often private/common land), planning consents, water regulation and protection of rural landscapes. Recent legislation (Minerals Security Taskforce, 2023) has streamlined some metal mining approvals, but still UK faces high taxation, legal, and capital costs compared to major mining jurisdictions. With silver near ~$25–30/oz, only very high-grade or large deposits could justify these hurdles. Given typical UK ore grades (<0.1% Ag) and the long lead-time (often >10 years) to develop a mine under modern permitting, most analysts view large-scale UK silver mining as uneconomic presently.

Viability Factors and Risks

Economic: Silver prices are volatile (recently ~$25–35/oz), making marginal projects risky. Because ~70% of supply is by-product【51†L169-L174】, silver-specific price spikes may not spur new mining quickly. Capital intensity is high; most modern silver projects require >100–200 kt ore/day mills and complex processing (cyanide or flotation), costing >$100 million capex. Operating costs vary: by-product recovery in a large zinc mine may cost $10–20/oz Ag (since Ag is credit), whereas a small high-grade silver mine (like Fresnillo) might see costs <$5/oz Ag. Projects must also contend with currency, energy, and input costs (diesel, labour) which have risen in many jurisdictions.

Regulatory/Environmental: Silver mining entails typical metal mining impacts (tailings, acid drainage risk, cyanide use). Countries vary: Chile and Canada have rigorous environmental review; Latin America and Asia can be less predictable (risks of permit delays or social conflict). In Europe (including UK), environmental and community opposition is high, so permitting is time-consuming and expensive. Modern tailings standards (e.g. dry-stack, monitoring) add cost but reduce risk. Interestingly, increased demand for silver in solar panels and electronics has prompted some jurisdictions to encourage domestic supply; for example, “Critical Minerals” policies in UK and Canada may offer incentives.

Logistics/Political: Top silver regions (Mexico, Peru, China) have good mining infrastructure (roads, power) but also risks (e.g. social protests in Peru, water scarcity). Remote areas (Alaska, Siberia, Sahara, Arctic) face higher costs (transport, energy) and seasonal constraints. Political risk is medium; silver miners operate in various regimes: from stable (Australia, USA) to more volatile (some Latin American, African countries). Global trade tensions have limited product exports rarely, since silver is not as geopolitically sensitive as rare earths or lithium. Supply chain: ~30% of silver is recycled (photovoltaics, electronics), which buffers mine output shortfalls.

Technical: Ore metallurgy can be complex. Silver often occurs with lead, copper or antimony, requiring multi-stage processing. Older mines may have left high-grade veins unmined due to ore accessibility or low metal prices (e.g. low-grade “stockwork” zones). Advances in heap leaching (for oxide ores), froth flotation, and electrowinning have reduced costs. Exploration has also improved: geophysical methods (IP surveys to locate sulphides, remote-sensing for alteration) and geochemical drones are in use to find blind silver deposits.

Ranking Risk: Generally, projects in stable jurisdictions (Australia, North America, Europe) have lower political risk but higher costs/regulation. Latin America and Central Asia offer rich deposits (e.g. Bolivia’s Potosí, Mexico’s Sierra Madre) but varying social/climate risks. Recent instability in Guinea and Myanmar impacted those jurisdictions’ mining sectors; similarly, inflationary/credit environments (2020s) have strained mining budgets.

Summary of Key Data

Country/ Region	Reserves (kt, USGS)	Prod’n (‘24, t)	Major Deposit Types	Permitting/Access
Mexico	~37 (low)	~8,000 【19†L3991-L3994】	High-grade epithermal Ag–Au veins; porphyry Cu–Ag (Carbonate-repl.)	Mining-friendly; long history (e.g. Fresnillo plc)

Country/ Region	Reserves (kt, USGS)	Prod'n ('24, t)	Major Deposit Types	Permitting/Access
Peru	110 【76†L52-L60】	~3,510 【19†L4001-L4002】	Cu–Zn–Ag polymetallic (Antamina), Au–Ag epithermal	Strong mining sector; permit 2–3 yrs
China	~67 (reported)	~3,424 【19†L4001-L4002】	Polymetallic (Pb–Zn–Cu), MVT, minor epithermals	Can be opaque data; mix of large state mines
Russia	92 【48†L52-L60】	~1,275 【19†L4001-L4002】	Polymetallic (Dukat), VMS/ skarns (Kyzyl)	Strategic supply (e.g. Polymetal); stable mining regs
Poland	59 【82†L52-L60】	~1,531 【19†L3991-L3994】	Cu–Ag porphyry (KGHM Legnica-Głogów)	EU regulations; KGHM JVs (Sierra Gorda etc)
USA	23 (est.)	~1,050 【19†L3993-L3995】	Epithermal veins (Nevada), Cu porphyry (Rochester)	Stringent regs; high costs (Alaska remote)
Australia	91 【78†L52-L60】	~1,045 【19†L4009-L4011】	Ag–Pb (Cannington), VMS (Mt. Isa region), Au–Ag epithermals	Mining-friendly; remote projects (WA, QLD)
UK	~~nil/unk.~~	0.63 【19†L3967-L3969】	Low-grade Ag in old Pb–Zn veins (Carboniferous)	Strict planning/ environment; uneconomic small targets

Table: Selected silver-producing countries – reserves (USGS 2025) and 2024 production (BGS)【76†L52-L60】【19†L3991-L3994】. Deposit type indicates typical geology (many Ag outputs are by-products of base-metal/ gold mines). "Permitting/Access" gives a qualitative sense of exploration/mining climate.

Overall, top-tier silver prospects remain in well-known belts: e.g. Mexico's epithermal veins, Andean polymetallic provinces, and mature base-metal camps in North America and Asia. Regions with less exploration but geological potential (e.g. African or Russian greenfields) could host new discoveries, albeit with higher risk. The UK, by contrast, is a low-viability region: the last major silver-related mine closed decades ago, and current projects face low grades and high non-technical barriers.

Data gaps and uncertainties: Reserve figures may be outdated or conservative (particularly for China and emerging producers). Grades at undeveloped projects are often proprietary until feasibility, so true "potential" is uncertain. Also, many silver resources lie deeper (under cover) or are sub-economic under today's prices. We have relied on USGS and BGS as primary data; independent industry sources (e.g. S&P, Metals Focus) might differ. Environmental/social variables are hard to quantify but increasingly decisive for project success.

Sources: World production and reserves from USGS Mineral Commodity Summaries and BGS data 【76†L52-L60】【19†L3991-L3994】. Deposit and mine examples from industry and literature 【51†L169-L174】【85†L297-L305】. UK historical/contextual details based on BGS and legacy mining records【19†L3967-L3969】.